Fundamentals of Knowledge Representation

Knowledge Representation (KR)

- Knowledge Representation (KR) is one of the cornerstones of **Knowledge-Based Artificial Intelligence (KBAI)**.
- It provides the formal structures through which intelligent systems **store, manipulate, and reason with knowledge**.

Knowledge?

- **Knowledge** can be defined as **information that has been organized, interpreted, and understood so that it can guide intelligent action**.
- In AI, knowledge is not merely raw data—it includes facts, rules, heuristics, and relationships that allow an agent to **draw inferences and make decisions**.

Data vs Information vs Knowledge:

- **Data**: Raw, unprocessed symbols (e.g., numbers, text).
- **Information**: Data organized meaningfully (e.g., "Temperature = 25°C").
- **Knowledge**: Information coupled with context and reasoning (e.g., "If Temperature > 30°C, turn on cooling system").

- Knowledge is the foundation of **intelligent behavior** because it allows AI systems to **reason beyond immediate sensor inputs**.

Types of Knowledge

- Knowledge can be categorized in several ways depending on **how it is acquired, represented, and used**.

(1) Declarative Knowledge (Knowing That)

- Represents **facts, concepts, and relationships**.
- Example: “All humans are mortal,” or “Water boils at 100°C.”
- Often stored as **facts, semantic networks, frames, or ontologies**.

(2) Procedural Knowledge (Knowing How)

- Represents **how to perform tasks** or **methods to achieve goals**.
- Example: “To solve a quadratic equation, apply the quadratic formula.”
- Stored in **production rules or procedural scripts**.

(3) Explicit vs Tacit Knowledge

- **Explicit Knowledge**: Clearly articulated, can be stored in a knowledge base.
  - Example: Legal regulations, encyclopedic knowledge.
- **Tacit Knowledge**: Difficult to formalize, often experiential or intuitive.
  - Example: Recognizing faces, driving a car.

(4) Domain-Specific vs Domain-Independent Knowledge

- **Domain-Specific**: Knowledge about a particular field (medicine, chemistry).
- **Domain-Independent**: General reasoning strategies (logical reasoning, search strategies).

Knowledge Representation Requirements

- Effective knowledge representation must satisfy several requirements to support reasoning:

  (1) **Expressiveness:**

  a. Must be able to **represent complex relationships and structures**.

  (2) **Inference Support:**

  a. Should allow **deriving new knowledge from existing knowledge** efficiently.

  (3) **Understandability:**

  a. Human experts should be able to **interpret and validate the representation**.

1. **Efficiency:**
   - Must support **fast retrieval, update, and reasoning**, especially for large-scale knowledge bases.
2. **Modularity & Scalability:**
   - Supports **adding new knowledge without rewriting existing knowledge**.
3. **Consistency & Soundness:**
   - Avoids **contradictions**; reasoning results should be logically correct.

- Designing knowledge representations involves balancing **expressiveness and efficiency**.

Knowledge Representation Trade-offs

Aspect	Trade-off
Expressiveness	Rich representations allow modeling complex concepts but may make inference computationally expensive.
Efficiency	Simple representations support fast reasoning but may lack detail.
Declarative vs Procedural	Declarative knowledge is easy to understand but may require more reasoning steps; procedural knowledge is efficient but harder to maintain or explain.
Generality vs Domain Specificity	General models apply widely but may be less precise; domain-specific models are accurate but less reusable.

Example:

- First-order logic is very expressive and supports sound reasoning but can be **computationally heavy** for large knowledge bases.
- Rule-based systems are **less expressive** but faster for specific domains.

Representation Techniques

- Several techniques are widely used in KBAI:

(1) Logic-Based Representations

- **Propositional Logic:** Represents facts as true/false statements.
  - Example: "It is raining → Carry an umbrella."
- **First-Order Predicate Logic:** Adds objects, relations, and quantifiers.
  - Example: $\forall x(\text{Human}(x) \rightarrow \text{Mortal}(x))$
- **Advantages:** Sound reasoning, supports automated theorem proving.
- **Limitations:** Complex and sometimes inefficient for large domains.

(2) Semantic Networks

- Graph structures where **nodes represent concepts** and **edges represent relationships**.
- Example: “Bird → has → Wings”
- Useful for **inheritance hierarchies** and **conceptual relationships**.

(3) Frames

- Struct-like structures representing **objects, attributes, and values**.
- Example: Frame for a “Car”: {type: vehicle, wheels: 4, fuel: gasoline}
- Supports **default values and inheritance**, useful in expert systems.

(4) Production Rules

- **Condition → Action** statements: IF condition THEN action.
- Example: IF Temperature > 30°C THEN TurnOn(AirConditioner)
- Efficient procedural knowledge representation; widely used in **expert systems**.

(5) Ontologies

- Define **formal concepts and relationships** for a domain.
- Example: Medical ontology links diseases, symptoms, and treatments.
- Essential for **semantic reasoning, natural language understanding, and knowledge integration**.

(6) Hybrid Approaches

- Combine **logic, frames, rules, and probabilistic reasoning**.
- Example: Medical expert systems may use **rules for diagnosis and frames for patient data**.

Knowledge Base and Inference Engine:

- A **Knowledge-Based System (KBS)** typically has two main components:

  (1) Knowledge Base (KB)

  - Stores **domain knowledge** using KR techniques.
  - Contains facts, rules, frames, ontologies, or production rules.
  - Must be **consistent, up-to-date, and structured for reasoning**.

  (2) Inference Engine

  - **Reasoning component** that draws new knowledge from the KB.
  - Uses **forward chaining (data-driven)** or **backward chaining (goal-driven)**.

- Example:
  - **Forward chaining:** “Detect symptoms → Suggest possible diseases.”
  - **Backward chaining:** “Is this patient diabetic? → Check symptoms and test results.”

Integration:

- Together, KB + Inference Engine = **Expert System / KBAI system**
- Explanation facility may be added to **justify reasoning**, enhancing transparency.

Summary:

Knowledge Representation is the **backbone of intelligent systems**. A robust KR enables an AI system to:

- Store and structure domain knowledge efficiently.
- Reason logically and make informed decisions.
- Adapt to new information through non-monotonic and utility-based reasoning.
- Provide explanations and transparency for human users.

Key Points:

1. Knowledge goes beyond data—it must support **reasoning and action**.
2. Different types of knowledge require **different representation techniques**.
3. KR design involves balancing **expressiveness, efficiency, and maintainability**.
4. Modern AI systems often use **hybrid KR approaches** combining logic, frames, rules, and ontologies.
5. The combination of **Knowledge Base + Inference Engine** is central to KBAI success.